INCH-POUND

MIL-S-19500/608

22 April 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY,
POWER RECTIFIER, COMMON CATHODE OR COMMON ANODE CENTER TAP,
TYPES 1N6660 AND 1N6660R
JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon, schottky, power rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (TO-254AA isolated).

1.3 Maximum ratings.

Type	V_R 1/	V_{RWM} 1/	I_O 1/ 2/ 3/ $T_J = T_C = +100°C$	I_{FSM} 1/ $T_C = +25°C$, $t_p = 8.3$ ms	C_J 1/ at 5 V	T_{STG} and T_J
	V	V	A dc	A (pk)	pf	°C
1N6660, 1N6660R	45	45	15 A	300 A	2000	-65 to +150

1/ Each individual diode.
2/ Derate linearly at 300 mA/°C from $T_J = T_C = +100°C$ to +150°C. 300 mA/°C times 50°C = 15 A, the device rating.
3/ Total package current is limited to 30 A dc.

1.4 Primary electrical characteristics. $R_{\theta JC}$ = 1.65°C/W maximum, $R_{\theta JA}$ = 40°C/W maximum, each individual diode.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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ing shall be in accordance with table II of 4.5 Screening (JANS, JANTX, and JANTXV levels only). Screeni

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90% AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The US Government preferred system of measurement is the meter SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Lead formation and finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition requirements (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14, of MIL-S-19500.

3.3.2 Polarity. Polarity and terminal configuration shall be in accordance with figure 1 herein.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and 4.3.2.1 herein.

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4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c 1/	Method 3101 (see 4.3.2), Peak reverse energy test (see 4.3.3)	Method 3101 (see 4.3.2), Peak reverse energy test (see 4.3.3)
9 & 10	Not applicable	Not applicable
11	V_{F2} and I_{R1}	V_{F2} and I_{R1}
12	See 4.3.1 t = 240 hours	See 4.3.1 t = 48 hours
13	Subgroups 2 and 3 of table I herein; ΔV_{F2} = ±50 mV; ΔI_{R1} = 100% of initial value or ±250 µA dc, whichever is greater.	Subgroup 2 of table I herein; ΔV_{F2} = ±50 mV; ΔI_{R1} = 100% of initial value or ±250 µA dc, whichever is greater.

1/ Shall be performed anytime after screen 3.

4.3.1 Power Burn-in conditions. Burn-in conditions are as follows:

MIL-STD-750 method 1038, test condition A. T_C = +125°C; V_R = 36 V dc.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. Test each die separately. The maximum limit and conditions for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots will be used to establish a fixed screening limit (not to exceed the group A limit). Once a fixed limit has been established, monitor all future sealing lots using a sample from each lot to be plotted on the applicable SPC chart.

4.3.2.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750. Method 3101 (read and record date $Z_{\theta JX}$) derived conditions limits and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

4.3.3 Peak reverse energy test. The peak reverse energy test is to be performed using the circuit as shown on figure 2 or equivalent. The Schottky rectifier under test must be capable of absorbing the reverse energy, as defined, below:

I_{RM} = 2 A minimum, V_{RSM} = 54 V minimum, L = 260 µH

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_H measure current - - - - - - - - - - - - - - - 10 mA

b. I_H forward heating current - - - - - - - - - - - 15 - 50 A

c. t_H heating time - - - - - - - - - - - - - - - - - 50 ms

d. t_{MS} measurement delay time - - - - - - - - - - - 100 μs minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$(max) = 1.55°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 4: Condition for intermittent operation life for each diode are as follows:

I_F or I_O = 2 A (minimum); $\Delta T_J \geq 85°C$, +15°C, -5°C for 2,000 cycles minimum.

b. Subgroups 5: Condition for steady-state operation life (accelerated) is as follows:

Method 1038, condition A, $T_J = T_C$ = +150°C, V_R = 36 V dc.

c. Subgroup 6: Limit for thermal resistance is $R_{\theta JC}$ = 1.65°C/W maximum for each die.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 3: Condition for intermittent operation life for each diode are as follows:

I_F or I_O = 2 A (minimum); $\Delta T_J \geq 85°C$, +15°C, -5°C for 2000 cycles minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table III of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength are as follows:

Tension: Test condition A, weight = 4.5 kg (10 lbs), t = 15 seconds.

b. Subgroup 3: Condition for constant acceleration are as follows:

X_1, Y_1, Z_1, and Z_2 axis.

c. Subgroup 6: I_F or I_O = 2 A (minimum); $\Delta T_J \geq 85°C$, +15°C -5°C for 6000 cycles minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection. 1/ 2/

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 3/	3101	See 4.4.1	$Z_{\theta JX}$		1.55	°C/W
Forward voltage	4011	I_F = 5 A (pk) pulsed (see 4.5.1)	V_{F1}		0.55	V dc
		I_F = 15 A (pk) pulsed (see 4.5.1)	V_{F2}		0.75	V dc
		I_F = 30 A (pk) pulsed (see 4.5.1)	V_{F3}		1.0	V dc
Reverse current	4016	DC method, V_R = 45 V dc, (See 4.5.1)	I_{R1}		1.0	mA dc
Subgroup 3						
High temperature operation		T_A = +125°C				
Reverse current leakage	4016	DC method, pulsed (see 4.5.1)				
		V_R = 45 V (pk)	I_{R2}		40	mA dc
Low temperature operation		T_A = -55°C				
Forward voltage	4011	Pulsed (see 4.5.1) I_F = 15 A (pk)	V_{F4}		0.8	V dc
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc; f = 1 MHz V_{SIG} = 50 mV (p-p) (max)	C_j		2000	pF
Subgroup 5						
Dielectric withstand	1016	V_R = 500 V dc; All leads shorted; V measure from leads to case; Sample is n = 116, c = 0.	DWV		10	µA
Subgroup 6 & 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.
2/ Each individual diode.
3/ If 4.4.1 test conditions are performed in 100% screening, this test need not be performed in Group A.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			38 devices, c = 1
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071		
Electrical measurements		See table III, steps 1, 2, and 5	
Subgroup 2			38 devices, c = 1
Steady-state blocking life	1048	t = 1,000 hours; T_C = +125°C; V_R = 36	
Electrical measurements		See table III, steps 1 and 2	
Subgroup 3			
Not Applicable			
Subgroup 4			10 devices, c = 0
Thermal resistance	3101	See 4.5.2, $R_{\theta JC}$ = 1.65°C/W	
Subgroup 5 1/			22 devices, c = 0
Surge	4066	Condition B; T_A = +25°C; I_{FSM} = 300 A, 100 surges	
Electrical measurements		See table III, steps 1 and 2,	

1/ For each diode.

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TABLE III. Groups A, B, C, and E electrical end-point measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 15 A (pk), Pulsed (see 4.5.1)	V_{F2}		0.75	V dc
2.	Reverse current leakage	4016	V_R = 45 V dc, Pulsed (see 4.5.1) DC method	I_{R1}		1.0	mA dc
3.	Forward voltage	4011	I_F = 15 A (pk), Pulsed (see 4.5.1)	ΔV_{F2}	±50 mV dc from initial reading.		
4.	Reverse current	4016	V_R = 45 V dc, Pulsed (see 4.5.1) DC method	ΔI_{R1}	±250 µA dc or 100 percent from initial reading, whichever is greater.		
5.	Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		1.55	°C/W

1/ Each individual diode.
2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1 and 2.

b. Subgroup 4, see table III herein, steps 1, 2, 3, 4, and 5.

c. Subgroup 5, see table III herein, steps 1, 2, 3, and 4.

3/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1 and 2.

b. Subgroup 3, see table III herein, steps 1, 2, and 5.

c. Subgroup 6, see table III herein, steps 1 and 2.

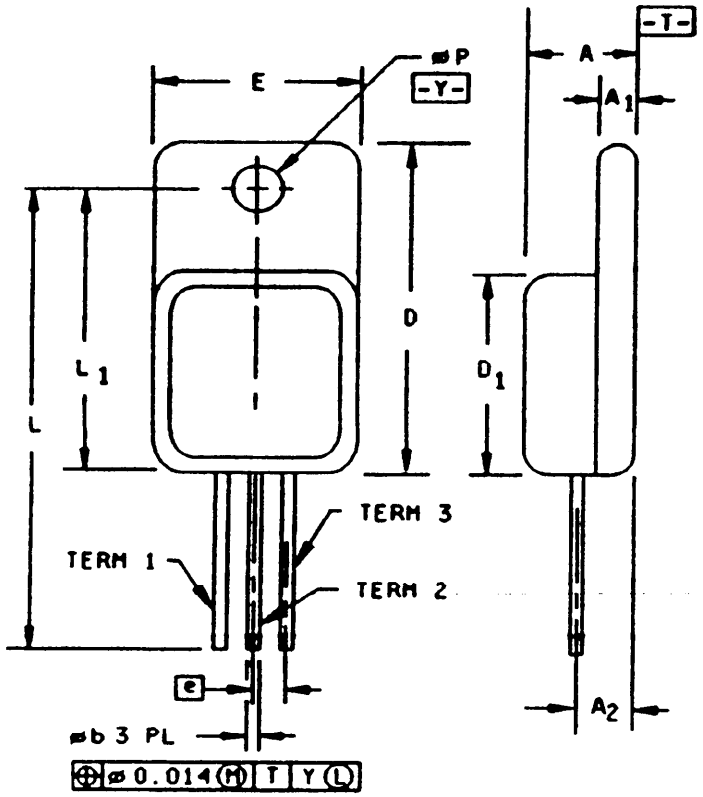
4/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table III herein, steps 1 and 2 for all levels.

b. Subgroup 6, see table III herein, steps 1, 2, and 5 for all levels.

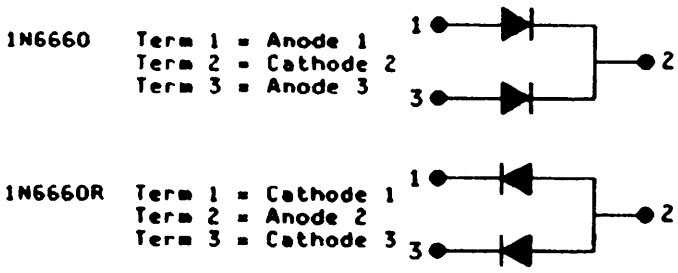
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Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	6.32	6.60	.249	.260
A_1	1 02	1.27	.040	.050
A_2	3.81		.150 BSC	
D	20.07	20.32	.790	.800
D_1	13.59	13.89	.535	.545
e	3.81		.150 BSC	
E	13.59	13.89	.535	.545
L	30.35	31.37	1.195	1.235
L_1	16.89	17.40	.665	.685
⌀P	3.53	3.78	.139	.149
⌀b	0.89	1.43	.035	.045

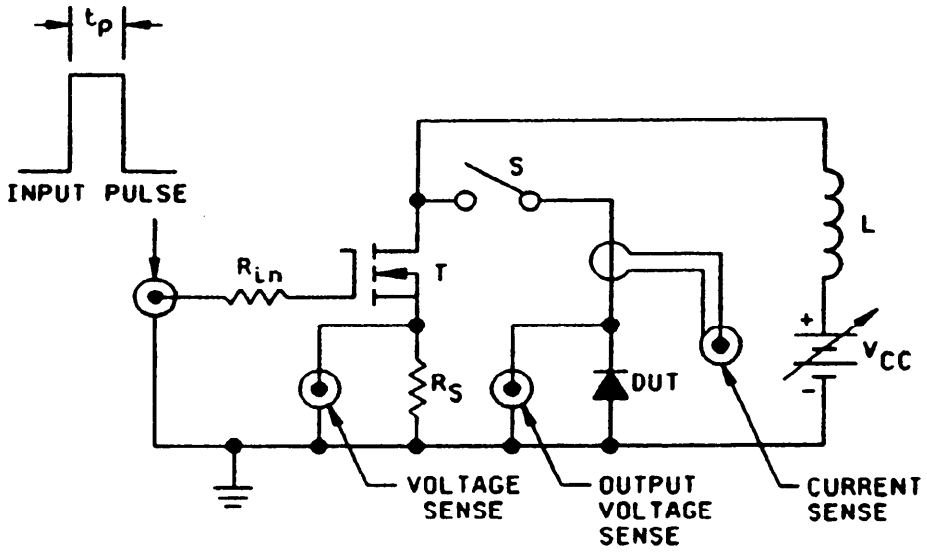
SCHEMATIC



NOTES:
1. Dimensions are in millimeters.
2. Inch-pound equivalents are given for general information only.
3. All terminals are isolated from case.

FIGURE 1. Physical dimensions and configuration (TO-254AA).

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Input pulse
V_G = 10 Volts
R_G = 50 ohms
P.W. = 30 μs
Duty cycle ≤ 1%

Rin = 50 ohms, 1 watt
R_S = 0.1 ohms, 1 watt
V_{CC} = 10 volts
L = 260 μH
T = IRF130/2N6756 or equivalent

PROCEDURES:
1. With S open, adjust pulse width to test current of 2 amps across R_S.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3)

FIGURE 2. Peak reverse energy test circuit.

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4.5.2 Thermal resistance. Thermal resistance shall be measured as follow in accordance with method 3101 or 5081. Each diode leg shall be measured.

a. I_M - - - - - - - - - - - - 10 mA

b. I_H - - - - - - - - - - - - 15 - 50 A

c. t_{md} - - - - - - - - - - - - 100 µs maximum

d. $R_{\theta JC}$ - - - - - - - - - - - - 1.65°C/W

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation and finish may be specified (see 3.3.1).

c. Product assurance level and type designator.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activity:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13

Preparing activity:
Navy - EC

Agent:
DLA - ES

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